

# A DIVISIBILITY OBSTRUCTION FOR JOINING TWO MOSER LATTICES

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**ABSTRACT.** Write  $R$  for the Moser lattice and  $\theta_a$  for the rotation with  $\cos \theta_a = (2a-1)/(2a)$ , so that  $a$  is the squared radius at which  $\theta_a$  pairs a point with its own image. We prove a divisibility obstruction: if the joint lattice  $R + \theta_a(R)$  has rank eight and  $4 \nmid a$ , it carries a 4-colouring by a group homomorphism, so no subset of it is 5-chromatic at any size, and no 5-chromatic union  $L \cup \theta_a(S)$  exists for any  $L, S \subseteq R$ . The restriction to  $R$  is real and not a convenience:  $R$  is one member of a nested family of lattices (Section 8), and Heule’s 553-vertex part already lives in the next one up, so a join built there is outside everything below. The obstruction is decidable in closed form at every rational  $a$ , before any graph is built; it kills every *rank-eight*  $a$  with  $4 \nmid a$  and is silent otherwise—the rank-four exceptions of Proposition 4 are untouched, as is every  $a$  with  $4 \mid a$ .

Its engine is a structural fact:  $3R$  is the maximal order of  $\mathbb{Q}(\sqrt{33}, i\sqrt{3})$ . From that,  $R$ ’s two integral geometric 4-colourings are the reductions modulo the two primes above 2, and occupancy of the shell at radius  $\sqrt{a}$  becomes a relative-norm condition, which the arithmetic of  $\mathcal{O}_V$  solves in closed form:  $R$  has a point at squared radius  $a = n/b$  in lowest terms if and only if  $b \in \{1, 3, 9\}$  and  $v_p(n)$  is even at every prime that splits in  $\mathbb{Q}(\sqrt{33})$  and stays inert in  $\mathbb{Q}(\sqrt{-3})$ . The rotations surviving both conditions have natural density zero among the integers.

Against that, constructions. Chaining a monochromatic-pair gadget of Parts’ gives a 5-chromatic join at  $a = 64n^2/9$  for every  $n$ , an infinite family that includes the one integer spindle our filters cannot decide. Seven spindles are verified to join, three of them— $\theta_{16}$ ,  $\theta_{28}$ ,  $\theta_{36}$ —by constructions for which we have found no precedent; each of the seven ships a DRAT certificate, checked, so that the unions are not 4-colourable is a proof and not a solver’s word. A periodic refinement kills eleven further integer candidates below 100. The interface any such construction must contain is treated in a companion paper and is not claimed here. The record of 509 is untouched.

## 1. INTRODUCTION

The Hadwiger–Nelson problem asks for  $\chi(\mathbb{R}^2)$ , the least number of colours needed so that no two points at distance 1 share one. De Grey [1] proved  $\chi(\mathbb{R}^2) \geq 5$  in 2018 with a 1581-vertex unit-distance graph; Polymath16, Heule [4] and Parts [6] drove the vertex count down to 509, where it has stood since 2020.

The record is built out of one lattice, and so is everything here. Fix the *Moser lattice*  $R$ , the subgroup of  $(\mathbb{C}, +)$  generated by the 30 unit vectors

$$\exp(i(k \cdot 60^\circ + j \theta_3/2)), \quad 0 \leq k < 6, \quad |j| \leq 2, \quad \theta_3 = \arccos \frac{5}{6},$$

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*Date:* September 20, 2026.

*2020 Mathematics Subject Classification.* 05C15, 52C10, 05C62, 11R11.

*Key words and phrases.* Hadwiger–Nelson problem, unit-distance graph, chromatic number of the plane, Moser lattice, real quadratic field.

the half-angle  $\varphi = e^{i\theta_3/2} = \frac{\sqrt{33}}{6} + \frac{\sqrt{3}}{6}i$  being needed for the twelve with  $j$  odd. It is dense in the plane, and it is a *group* and not a ring—of the 900 products of pairs of unit vectors only 684 lie in it. But  $3R$  is a ring, and is the maximal order of the field  $R$  spans:

$$R = \frac{1}{3} \mathcal{O}_V, \quad V = \mathbb{Q}(\sqrt{33}, i\sqrt{3}),$$

which is Proposition 8 and is what the rest of the paper runs on. The *Moser ring*  $R$  generates is strictly larger, with denominators  $3^k$ ; Gibbs' homomorphism and Hubai's 22 colourings [5] live there, and everything below is about  $R$  itself.

Parts [6] classifies the record's constructions into four types, of which *type M*—two copies of a base graph rotated relative to each other and glued at a shared origin,

$$G = L \cup \theta(S), \quad L, S \subseteq R,$$

—“is of the greatest interest, because so far it gives 5-chromatic graphs with the least number of vertices”. The only rotation he records for it is  $\theta_4 = \arccos(7/8)$ ; writing in 2020 he could not have had the four that [9] found the year after. Of the family he adds:

“Working constructions of type M with other rotations are not known yet.”

That sentence is the question this paper takes up.

*Two words, used throughout.* A rotation  $\theta$  **joins** when there are finite  $L, S \subseteq R$  for which  $L \cup \theta(S)$ , glued at the shared origin, is not 4-colourable—equivalently, by Lemma 21, when some such union contains an induced subjoin of chromatic number exactly 5. It is a property of the rotation and the lattice, not of any pair of halves. A rotation is a **spindle** when its cosine is rational; writing that cosine as  $(2a-1)/(2a)$  names it  $\theta_a$ , and  $\sqrt{a}$  is its *shell* radius, the one radius at which a point and its image are a unit apart. Section 2 fixes the normalisation that makes  $a$  well defined;  $\theta_4$  is a spindle and is Heule's rotation.

**Theorem 1** (the divisibility law). *Let  $a$  be a positive rational and suppose the joint lattice  $\Lambda_a = R + \theta_a(R)$  has rank 8—which by Proposition 4 is every  $a$  except those with  $4a-1 = 3k^2$  or  $11k^2$  for a rational  $k$ . If  $4 \nmid a$  then  $\Lambda_a$  admits a homomorphism 4-colouring. If  $4 \mid a$  and the shell of  $R$  at radius  $\sqrt{a}$  is non-empty, it admits none.*

A homomorphism 4-colouring properly colours the *entire* infinite lattice, at every radius, so:

**Corollary 2.** *If  $\Lambda_a$  has rank 8 and  $4 \nmid a$ , no subset of  $\Lambda_a$  is 5-chromatic; in particular no 5-chromatic  $L \cup \theta_a(S)$  exists, at any size.*

The shell hypothesis carries only the second half, and it is sufficient rather than necessary: what the proof needs is a vector of length  $\frac{1}{2}$  in  $\Lambda_a$ , which an occupied shell always supplies and an empty one may supply by another route— $\Lambda_{24}$  has twelve, exhibited in Section 3.

**What is proved here.** The engine is arithmetic, and it is the part likeliest to outlast the obstruction. Proposition 8 identifies  $3R$  as  $\mathcal{O}_V$ ; from it, Theorem 9 shows that  $R$ 's two integral geometric 4-colourings are reduction modulo the two primes above 2—a count due to Dúcz, here given a structural reason—and Theorem 18 answers in closed form which rationals are squared distances in  $R$ :  $a = n/b$  in lowest terms is one exactly when  $b \in \{1, 3, 9\}$  and  $v_p(n)$  is even at every prime splitting in

$\mathbb{Q}(\sqrt{33})$  and inert in  $\mathbb{Q}(\sqrt{-3})$ . Corollary 19 makes the surviving integer rotations a set of density zero. Proposition 3 removes the ball from the whole question: a cross edge is a unit vector of  $\Lambda_a$ , of which there are finitely many, exactly enumerable.

Theorem 1 is the application. It is one-sided and its reach is bounded in a way worth stating at once: it kills three rational  $a$  in four, and its own second half proves that at  $4 \mid a$  with an occupied shell no homomorphism colouring exists at all—so the method is silent exactly where the working rotations live. It is silent outside the spindle family too, four of the five rotations known to work having irrational cosine. Section 7 kills eleven further integer candidates by a periodic refinement, and Section 6 exhibits seven joining spindles, each with a DRAT certificate a referee can check; Lemma 20 chains a gadget of Parts' into a join at  $a = 64n^2/9$  for every  $n$ , so infinitely many spindles join. The record of 509 is untouched, and nothing here brings it closer.

The restriction to  $R$  is real.  $R$  is one member of a nested family (Section 8), and Heule's 553-vertex part already lives in the next one up, so a join built there is outside everything proved below.

**What is not new.** The two geometric 4-colourings of  $R$  and the fact that there are exactly two are Dúcz's, after Hubai; his proof is a sublattice argument on coordinates mod 2, and contains no prime ideal and no number field, and his lattice is the index-9 sublattice generated by  $1, \omega_1, \omega_3, \omega_1\omega_3$ , so his count and Theorem 9 are about different groups. Parts already writes his vertices over  $\sqrt{33}$ , subject to a congruence that normalises coordinates inside one fixed lattice; the divisibility here is a condition on the *rotation*. The five previously known type-M rotations are Heule's and the four of [9]. What is claimed below is the arithmetic, the law, and the constructions of Section 6.

On that claim we can be specific rather than merely careful. The four papers that work in or on this lattice—[2], [6], [9] and [3]—contain, between them, no occurrence of *maximal order*, *ring of integers*, *prime ideal* or *number field*; we checked their full texts rather than their abstracts. The squared-distance spectrum that is classical here is the triangular lattice's, where the attainable values are the Loeschian numbers  $a^2 + ab + b^2$ ; we have found no statement, in those papers or in the Polymath16 material, of the corresponding set for  $R$ , which is rank 4 and whose spectrum is Theorem 18. That is evidence and not proof: this literature is partly distributed across blog comments and data files, and a negative over it cannot be certified.

## 2. THE JOINT LATTICE, AND WHY NO BALL IS NEEDED

**Proposition 3.** *Suppose  $\Lambda_a = R \oplus \theta_a(R)$  has rank 8. Then  $(p, \theta_a(q)) \mapsto p - \theta_a(q)$  is a bijection from the cross edges of  $R \cup \theta_a(R)$ —pairs  $p \in R$  and  $\theta_a(q) \in \theta_a(R)$  at distance 1 from each other—onto the unit vectors of  $\Lambda_a$ . There are finitely many, though the union is dense. Moreover such an edge is monochromatic under a homomorphism  $\varphi$  of  $\Lambda_a$  exactly when  $\varphi(u) = 0$ , so what a cross edge contributes depends only on  $u$ .*

*Proof.* If  $|p - \theta_a(q)| = 1$  then  $u = p - \theta_a(q)$  is a unit vector of  $\Lambda_a$ . Conversely the direct sum writes a unit vector uniquely as  $u = r + \theta_a(s)$ , and then  $p = r$ ,  $q = -s$  is the only pair with  $p - \theta_a(q) = u$ . Writing  $\varphi = (\varphi_1, \varphi_2)$  on the two summands,

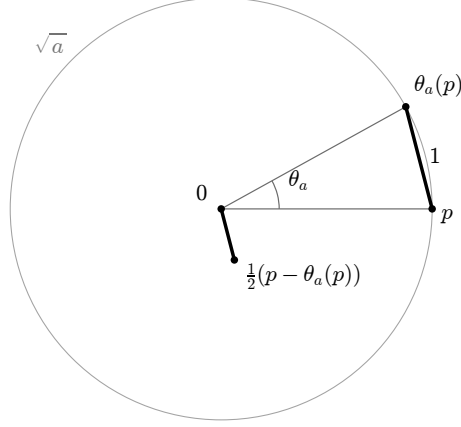


FIGURE 1. The spindle at squared radius  $a$ . Two shell points at distance  $\sqrt{a}$  are carried onto each other by  $\theta_a$  and land at distance exactly 1, which is what the rotation is for. Half that displacement has length  $\frac{1}{2}$  always, since  $|1 - \theta_a| = 1/\sqrt{a}$ ; whether it lies in the lattice is the whole content of Theorem 1. Drawn for  $a = 4$ ,  $\cos \theta_4 = 7/8$ .

$\varphi(u) = \varphi_1(p) - \varphi_2(\theta_a(q))$ , which vanishes exactly when the two ends receive the same colour.  $\square$

**Proposition 4.** *Let  $V \subset \mathbb{C}$  be the  $\mathbb{Q}$ -span of  $R$ . Then  $V = \mathbb{Q}(\sqrt{33}, i\sqrt{3})$ , a field of degree 4 over  $\mathbb{Q}$ , and*

$$\theta_a \in V \iff 4a - 1 = 3k^2 \text{ or } 11k^2 \text{ for a rational } k.$$

*If  $\theta_a \in V$  then  $\Lambda_a = R + \theta_a(R)$  has rank 4; otherwise it has rank exactly 8 and Theorem 1 applies. No intermediate rank occurs.*

**One symmetry, and the normalisation it forces.**  $R$  is carried to itself by complex conjugation and by  $-1$ , so for any  $L, S \subseteq R$ ,

$$\overline{L \cup \theta_a(S)} = \overline{L} \cup \theta_{a'}(-\overline{S}), \quad a' = \frac{a}{4a - 1},$$

with  $\overline{L}$  and  $-\overline{S}$  again inside  $R$ . The identity behind it is  $\theta_{a'} = -\overline{\theta_a}$ , which one reads off  $\cos \theta_{a'} = -\cos \theta_a$  and  $\sin \theta_{a'} = \sin \theta_a$ . Mirror images are isometric, so  $\theta_a$  joins if and only if  $\theta_{a'}$  does. The map  $a \mapsto a/(4a - 1)$  is an involution exchanging  $(\frac{1}{4}, \frac{1}{2})$  with  $(\frac{1}{2}, \infty)$  and fixing  $\frac{1}{2}$ , and it leaves  $v_2(a)$  alone whenever the denominator of  $a$  is odd.

We normalise to  $\cos \theta_a \geq 0$ , that is to  $a \geq \frac{1}{2}$ , and every statement below about which  $\theta_a$  join is a statement about such representatives. This is not cosmetic. It is the shell that breaks the symmetry:  $\theta_4$  spins a shell of 30 points, while its partner  $\theta_{4/15}$ —which carries the mirror image of the very same 509-vertex graph—has an empty shell, since  $15 \notin \{1, 3, 9\}$ . Un-normalised, a criterion phrased through shells would admit one of the pair and reject the other.

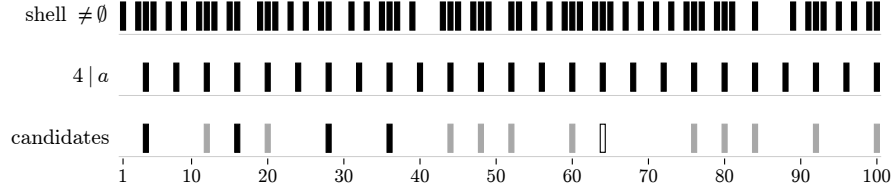


FIGURE 2. Theorem 1 as a sieve, over the integer spindles  $a \leq 100$ . Top: the 59 *positive* squared radii at which  $R$  has points, decided by the form (3) rather than by a ball, and in closed form by Theorem 18. Middle: the multiples of 4. Bottom: their intersection, the sixteen multiples of 4 with an *occupied* shell—the candidates, in the sense Section 6 fixes. The nine gaps there are the multiples of 4 whose shell is empty,  $a = 8, 24, 32, 40, 56, 68, 72, 88, 96$ ; they lie outside this accounting altogether, because at an empty shell Theorem 1 is silent in both directions. The law is a kill, not a construction, so the bottom row also carries what became of each candidate: filled, the four given a verified 5-chromatic union in Section 6; grey, the eleven the stronger filter of Section 7 kills; open,  $a = 64$ , on which that filter degenerates at every modulus tried—it joins all the same, by Lemma 20, which builds a graph there rather than testing one.

Then  $|p - \theta_a(p)| = |p||1 - \theta_a| = |p|/\sqrt{a}$ , so a point and its own image are a unit pair exactly when  $|p| = \sqrt{a}$ : the rotation's own pairing lives on that *shell* and nowhere else, and Figure 1 is the whole mechanism in one picture. Other points do acquire cross edges—to images of *other* points, as Figure 3 shows—but if the shell is empty the rotation has nothing of its own to act on.

Voronov, Neopryatnaya and Dergachev [9] enumerate rotations carrying one subset of vertices onto another; their first series runs on  $R$  and reports five that work, Heule's among them, so five working type-M rotations were known and not one. They are one orbit: all five are  $\rho u$  with  $\rho = \theta_4$  and  $u$  a unit vector of  $R$ , and exactly one of the five— $\rho$  itself—has rational cosine. The other four are therefore not spindles, and in the family classified here exactly one working member was known.

Every earlier test of a rotation built a ball of lattice points, found its unit edges and ran a SAT solver. That has a free parameter—the ball—and a negative answer about a ball is not an answer about the lattice: a rotation whose shell at radius  $\sqrt{a}$  lies outside the ball is being dismissed on a graph that never contained a single pair  $p, \theta_a(p)$ . Section 6 exhibits six rotations that a ball at the record's own scale misses for exactly that reason.

No ball is needed. Write  $W$  for the 30 unit vectors of  $R$ . For a candidate  $\theta$ , both halves of  $L \cup \theta(S)$  lie in

$$\Lambda_\theta = \langle W \cup \theta(W) \rangle,$$

the group they and their images generate: a finitely generated subgroup of  $(\mathbb{C}, +)$  with no radius in it. Its unit vectors are enumerable exactly:  $q(z) = \text{Tr}(z\bar{z})$  is a positive-definite quadratic form on  $\Lambda_\theta \otimes \mathbb{Q}$ , and a unit vector has  $z\bar{z} = 1$  and hence

$q(z) = \text{Tr}(1)$ , so the unit vectors lie *among* the integer points of one ellipsoid. That set is finite and can be listed exactly; sifting it by the exact condition  $z\bar{z} = 1$ —which is four rational conditions where  $q$  is one, or one condition over  $F$  where  $q$  is a single rational—leaves the unit vectors, a complete list obtained without reference to any ball.

**The criterion.** Look for a group homomorphism  $\varphi : \Lambda_\theta \rightarrow \mathcal{A}$  with  $|\mathcal{A}| = 4$  and  $\varphi(u) \neq 0$  for every unit vector  $u$ . If one exists then for any  $p, q \in \Lambda_\theta$  at distance 1 the difference  $p - q$  is a unit vector, so  $\varphi(p) \neq \varphi(q)$  and  $p \mapsto \varphi(p)$  is a proper colouring of the whole infinite lattice. Hence

$$\varphi \text{ exists} \implies \text{no subset of } \Lambda_\theta \text{ is 5-chromatic, ever.}$$

The converse fails—a lattice may be 4-colourable by a colouring that is not a homomorphism—so the test is a *one-sided kill*, and a survivor is a candidate and nothing more. There are two groups of order 4, the Klein group and  $\mathbb{Z}/4$ , and both must be ruled out.

A one-sided kill has one way to be wrong: killing something that works. Every rotation known to carry a 5-chromatic type-M graph is therefore a test of the criterion, and there are five—Heule’s and the four of [9]. Run against all five, the criterion returns *unknown* each time and kills none; the five joint lattices have rank 8 and carry 126, 108, 108, 90 and 90 unit vectors respectively. That is the only outcome consistent with their being 5-chromatic.

Four of the five are informative and one is not. Heule’s rotation is  $\theta_4$ , and  $4 = 4 \cdot 1^2$ , so Proposition 5 forces its survival on any lattice closed under  $\omega$ :  $\Lambda_4$  holds a half-unit vector and the criterion could not have killed it whatever the implementation did. The test is therefore a test on the other four, where nothing forces the outcome.

### 3. THE CRITERION IN CLOSED FORM

**Proposition 5.** *Let  $\Lambda$  be any subgroup of  $(\mathbb{C}, +)$  closed under  $\omega = e^{i\pi/3}$  and containing a unit vector  $g$ , and let  $a = 4m^2$  for a positive integer  $m$ . Then  $\Lambda + \theta_a(\Lambda)$  contains a vector of length  $\frac{1}{2}$ , so by Proposition 6 the criterion cannot kill  $\theta_a$ : neither group of order 4 admits a homomorphism avoiding the unit vectors. Both lattices here are  $\omega$ -closed— $R$  by construction, and Haugland’s because  $\omega$  is one of its 42 roots of unity.*

*Proof.*  $|2mg| = 2m = \sqrt{a}$ , so  $2mg$  lies on the shell, and

$$h_g = \frac{1}{2}(2mg - \theta_a(2mg)) = m(1 - \theta_a)g$$

lies in  $\Lambda + \theta_a(\Lambda)$  with length  $m|1 - \theta_a| = m/\sqrt{a} = \frac{1}{2}$ . Proposition 6 does the rest, both halves of it, since  $\Lambda$  is  $\omega$ -closed.  $\square$

**Proposition 6.** *Suppose  $\Lambda_\theta$  contains a vector of length  $1/2$ . Then no homomorphism onto a group of order 4—Klein or cyclic—avoids every unit vector.*

*Proof.* Let  $|h| = 1/2$ , so  $2h$  is a unit vector of  $\Lambda_\theta$ . In the Klein group every element satisfies  $2x = 0$ , so  $\varphi(2h) = 2\varphi(h) = 0$  and the unit vector  $2h$  is killed.

In  $\mathbb{Z}/4$ ,  $\varphi(2h) = 2\varphi(h) \neq 0$  forces  $\varphi(h)$  odd, and that is already impossible. Write  $\omega = e^{i\pi/3}$ , a unit vector of  $R$  with  $\omega R = R$ , so  $\omega\Lambda_\theta = \Lambda_\theta$  and  $\omega h$ ,  $\omega^2 h$  are half-unit vectors alongside  $h$ . Since  $\omega^2 = \omega - 1$ ,

$$h + \omega h + \omega^2 h = 2\omega h \in 2\Lambda_\theta,$$

so  $\varphi(h) + \varphi(\omega h) + \varphi(\omega^2 h) = 2\varphi(\omega h)$  is even, while three odd values sum to an odd one.  $\square$

Both halves are unconditional, and the  $\omega$ -identity is why: it is a relation of *odd* weight among half-unit vectors, and it exists wherever one half-unit vector does. The sweep runs over every multiple of 4 up to  $a = 100$ , and those are the only integer spindles that can have a half-unit vector at all: where  $4 \nmid a$ , Theorem 1 supplies a homomorphism 4-colouring, which Proposition 6 forbids wherever one exists. Nineteen of them do have one—the sixteen with an occupied shell, where Theorem 1's own construction produces it, together with  $a = 24, 72$  and  $88$ , where the shell is empty and twelve appear regardless—and on every one of the nineteen the relation is found, as it must be.

Half-unit vectors are therefore the object to count. The shell supplies a family of them in closed form—though not, as  $a = 24$  will show below, all of them. For  $p$  on the shell,  $p$  and  $\theta_a(p)$  are both at distance  $\sqrt{a}$ , and

$$\frac{1}{2}(p - \theta_a(p))$$

has length exactly  $1/2$  for every  $a$ , since  $|1 - \theta_a| = 1/\sqrt{a}$ . Now  $p - \theta_a(p)$  always lies in  $\Lambda_a$ , being a difference of one element of  $R$  and one of  $\theta_a(R)$ ; whether *half* of it does is another matter, and that is the whole content of the classification:  $\frac{1}{2}(p - \theta_a(p)) \in \Lambda_a$  exactly when  $p - \theta_a(p) \in 2\Lambda_a$ .

That classifies the half-unit vectors *the shell produces*, and it is not a classification of all of them. At  $a = 24$  the shell is empty, so the construction above yields nothing at all—and yet  $\Lambda_{24}$  holds twelve vectors of length  $\frac{1}{2}$ . One of them needs no machinery to write down. Since  $R \cap \mathbb{Q} = \frac{1}{3}\mathbb{Z}$ , both  $r = -\frac{5}{3}$  and  $s = 2$  lie in  $R$ , and with  $\cos \theta_{24} = \frac{47}{48}$ ,  $\sin \theta_{24} = \frac{\sqrt{95}}{48}$ ,

$$r + \theta_{24}(s) = -\frac{5}{3} + 2\left(\frac{47}{48} + i\frac{\sqrt{95}}{48}\right) = \frac{7 + i\sqrt{95}}{24}, \quad \left|\frac{7 + i\sqrt{95}}{24}\right|^2 = \frac{49 + 95}{576} = \frac{1}{4}.$$

So Proposition 6 applies at  $a = 24$  and no homomorphism 4-colouring exists there, although Theorem 1 says nothing: its second half asks for an occupied shell, and an occupied shell is sufficient for a half-unit vector without being necessary. In the language of Step 1 below this is the case  $s = \lambda r$  with  $\lambda = -\frac{6}{5}$ , where  $|r|^2(a(1 + \lambda)^2 - \lambda)/a = \frac{1}{4}$ ; the shell is  $\lambda = -1$ , and it is not the only value that can reach length  $\frac{1}{2}$ . Section 7 takes this up again, where it explains survivals the law cannot.

#### 4. THE ARITHMETIC OF THE MOSER LATTICE

**4.1. A module over a real quadratic ring.** In the chart

$$z = \frac{A + B\sqrt{33}}{12} + i\frac{C\sqrt{3} + D\sqrt{11}}{12}, \quad A, B, C, D \in \mathbb{Z}, \quad (1)$$

the lattice  $R$  is cut out by the congruences

$$(i) \quad A \equiv B \equiv C \equiv D \pmod{2}, \quad (ii) \quad A + B + C + D \equiv 2A \pmod{4}, \quad (2)$$

sixteen of the 256 residue classes mod 4. (The chart without them is a group containing  $R$  with index 16, so reading a chart count as a lattice count overcounts—at rational squared radii at least; Lemma 7 below says it never does at integer ones.)

Squared lengths are

$$|z|^2 = \frac{(A^2 + 33B^2 + 3C^2 + 11D^2) + 2(AB + CD)\sqrt{33}}{144}, \quad (3)$$

so they lie in  $\mathbb{Q}(\sqrt{33})$ , and are *rational exactly when*  $AB + CD = 0$ , in which case they lie in  $\frac{1}{144}\mathbb{Z}$ . Write

$$S = A^2 + 33B^2 + 3C^2 + 11D^2$$

for the numerator, so a rational squared length is  $S/144$ . (This  $S$  is the form's value throughout Sections 4 and 5; the small half of a construction carries the same letter only outside them.)

At *integer* squared radii the congruences cost nothing, which is worth recording because it is what makes the chart safe to compute with there.

**Lemma 7.** *Let  $A, B, C, D \in \mathbb{Z}$  satisfy  $AB + CD = 0$  and  $16 \mid S$ . Then (2) holds automatically.*

*Proof.* Squares modulo 16 depend only on their bases modulo 8, so both hypotheses and both conclusions are determined by  $(A, B, C, D)$  modulo 8, and  $8^4 = 4096$  classes decide the statement. Of those, 320 satisfy  $16 \mid S$  while failing (2), and not one of the 320 has  $AB + CD \equiv 0 \pmod{8}$ .  $\square$

If  $|z|^2 = a$  is a positive integer then  $S = 144a$ , so  $16 \mid S$  and  $AB + CD = 0$ , and Lemma 7 applies: every integral chart point at an integer squared radius already lies in  $R$ . Measured, the two agree exactly—identical counts at every  $a \leq 300$ , not merely the same emptiness.

The module structure is in fact a great deal more than a module structure.

**Proposition 8.**  *$3R$  is the maximal order  $\mathcal{O}_V$  of  $V = \mathbb{Q}(\sqrt{33}, i\sqrt{3})$ ; that is,  $R = \frac{1}{3}\mathcal{O}_V$ . Writing  $F = \mathbb{Q}(\sqrt{33})$  for the real quadratic subfield and  $\bar{\phantom{x}}$  for complex conjugation, which generates  $\text{Gal}(V/F)$ ,*

$$|z|^2 = z\bar{z} = N_{V/F}(z),$$

*so a squared radius of  $R$  is one ninth of a relative norm of an algebraic integer.*

*Proof.* Write coordinates over the  $\mathbb{Q}$ -basis  $(1, \sqrt{33}, i\sqrt{3}, i\sqrt{11})$  of  $V$ . Reducing the 30 generators of  $R$  to Hermite form and multiplying by 3 gives a  $\mathbb{Z}$ -basis of  $3R$ :

$$b_1 = \frac{1}{4}(1, 1, 1, -1), \quad b_2 = \frac{1}{2}(0, 1, 0, -7), \quad b_3 = \frac{1}{2}(0, 0, 1, 1), \quad b_4 = (0, 0, 0, 1).$$

It contains 1, because  $\frac{1}{3} \in R$  by (1) at  $(4, 0, 0, 0)$ , and it is closed under multiplication: the sixteen products  $b_i b_j$  reduce to integer combinations of the  $b_k$  by the rules  $\sqrt{3}\sqrt{11} = \sqrt{33}$ ,  $\sqrt{3}\sqrt{33} = 3\sqrt{11}$  and  $\sqrt{11}\sqrt{33} = 11\sqrt{3}$ . So  $3R$  is an order in  $V$ . (The same closure holds on all 900 products of unit vectors, where  $R$  itself fails: only 684 of them return.)

For the discriminant,  $\text{Tr}_{V/\mathbb{Q}}(x) = 4x_1$  on these coordinates, since the four embeddings negate  $\sqrt{33}$  and  $i\sqrt{3}$  independently and kill every other basis vector. Hence

$$(\text{Tr}(b_i b_j)) = \begin{pmatrix} 5 & -22 & 4 & 11 \\ -22 & -506 & 77 & 154 \\ 4 & 77 & -14 & -22 \\ 11 & 154 & -22 & -44 \end{pmatrix}, \quad \det = 1089.$$

Now compare with the field. Put  $F = \mathbb{Q}(\sqrt{33})$ , so  $V = F(i\sqrt{3})$  and  $\text{disc}(F) = 33$  because  $33 \equiv 1 \pmod{4}$ . Two standard facts close the argument without computing  $\text{disc}(V)$  separately. First, for any order  $\mathcal{O}$  of  $V$ ,

$$\text{disc}(\mathcal{O}) = [\mathcal{O}_V : \mathcal{O}]^2 \text{disc}(V).$$

Second, the tower formula for relative discriminants, applied to  $V/F$  of degree 2 gives  $\text{disc}(V) = N_{F/\mathbb{Q}}(\mathfrak{d}_{V/F}) \text{disc}(F)^2$ , so  $\text{disc}(V)$  is a positive multiple of  $33^2 = 1089$ . Taking  $\mathcal{O} = 3R$  and substituting,

$$1089 = [\mathcal{O}_V : 3R]^2 \cdot N_{F/\mathbb{Q}}(\mathfrak{d}_{V/F}) \cdot 1089,$$

which forces both factors to be 1. So  $3R = \mathcal{O}_V$ , and as a by-product  $\mathfrak{d}_{V/F} = (1)$ :  $V/F$  is unramified at every finite prime, which is what Theorem 18 will use. Remark 17 reaches the same conclusion from genus theory, and identifies  $V$  as the narrow Hilbert class field of  $\mathbb{Q}(\sqrt{33})$ .

The norm identity is the definition of  $N_{V/F}$ : complex conjugation fixes  $\sqrt{33}$  and negates  $i\sqrt{3}$ , so it generates  $\text{Gal}(V/F)$  and  $z\bar{z} = N_{V/F}(z)$ .  $\square$

Two consequences follow, and one of them is the whole of Section 6's criterion.

- $R$  is a module over  $\mathcal{O} = \mathbb{Z}[(1 + \sqrt{33})/2]$ , the ring of integers of  $\mathbb{Q}(\sqrt{33})$ . This is a finite check and not a sample. The congruences (2) depend only on  $(A, B, C, D)$  modulo 4, and exactly 16 of the 256 classes satisfy them. Multiplication by  $\sqrt{33}$  sends  $(A, B, C, D)$  to  $(33B, A, 11D, 3C)$ —using  $\sqrt{3}\sqrt{33} = 3\sqrt{11}$  and  $\sqrt{11}\sqrt{33} = 11\sqrt{3}$ —and carries each of the 16 classes to one of the 16. For  $(1 + \sqrt{33})/2$  the same map, halved, is integral because  $A \equiv B$  and  $C \equiv D \pmod{2}$  on those classes, and again preserves them. Since  $\text{rank}_{\mathbb{Z}} R = 4$  and  $\text{rank}_{\mathbb{Z}} \mathcal{O} = 2$ , the lattice  $R$  has rank 2 over  $\mathcal{O}$ .
- $R$  is not a lattice in the geometric sense. It is dense, for the reason given in Section 1; its short vectors are as short as one likes, and a depth-4 ball already contains one with  $|v|^2 = (23 - 4\sqrt{33})/9 \approx 0.0024$ . Any argument wanting a smallest vector must restrict to *rational* norms, where  $\frac{1}{144}\mathbb{Z}$  supplies one.

The smallest rational squared norms of  $R$ , decided exactly from (3) together with (2), are

$$\frac{1}{9}, \frac{1}{3}, \frac{4}{9}, \frac{5}{9}, \frac{7}{9}, 1, \dots$$

so  $1/9$  is the least, attained by six points, and it is the only one below  $1/4$ . The value  $1/36$  is not attained at all, a fact used in Section 5.

**4.2. The two colourings are prime reductions.** Since  $33 \equiv 1 \pmod{8}$ , the prime 2 splits in  $\mathbb{Q}(\sqrt{33})$ :

$$2 = -\pi_1\pi_2, \quad \pi_1 = \frac{5 + \sqrt{33}}{2}, \quad \pi_2 = \frac{5 - \sqrt{33}}{2}, \quad N(\pi_i) = -2.$$

Because  $R$  is an  $\mathcal{O}$ -module,  $\pi_i R$  is a subgroup, and  $[R : \pi_i R] = N(\pi_i)^2 = 4$ , exactly the index a 4-colouring needs, where  $[R : 2R] = 16$  is four times too big.

**Theorem 9** (the colourings are prime reductions). *The rational prime 2 splits in  $\mathbb{Q}(\sqrt{33})$  as  $-\pi_1\pi_2$  with  $\pi_{1,2} = (5 \pm \sqrt{33})/2$ . The lattice  $R$  is a module over  $\mathcal{O} = \mathbb{Z}[(1 + \sqrt{33})/2]$ , and the two integral geometric 4-colourings of  $R$ —there are*

exactly two, as the proof below counts—are the reductions modulo  $\pi_1 R$  and  $\pi_2 R$ . In particular there are two of them because 2 splits into two primes.

*Proof of Theorem 9.* First, the quotient has no choice but to be Klein. By integrality the colour classes are the cosets of a subgroup  $H$  of index 4. By geometry the isometry  $\omega = e^{i\pi/3}$  of  $R$  permutes those classes, and it fixes the origin, so it fixes the class containing it:  $\omega H = H$ , and  $\omega$  induces an automorphism  $T$  of  $R/H$ . Since  $\omega^2 = \omega - 1$ ,

$$T^2 - T + I = 0.$$

If  $R/H$  were cyclic of order 4 then  $T = \pm I$ , the only automorphisms it has, and the left side would be  $I$  or  $3I$ —neither zero in  $\text{End}(\mathbb{Z}/4)$ . So  $R/H$  is the Klein group,  $2R \subseteq H$ , and the search may be confined to subgroups containing  $2R$ .

That done, a homomorphism onto the Klein group killing no unit vector is the same thing as a subgroup of index 4 containing  $2R$  and meeting none of the 30 unit vectors, so the whole question lives in  $R/2R$ —and Proposition 8 says exactly what that quotient is. Multiplication by 3 carries  $R$  isomorphically onto  $\mathcal{O}_V$  and  $\pi_i R$  onto  $\pi_i \mathcal{O}_V$ , so  $R/\pi_i R = \mathcal{O}_V/\pi_i \mathcal{O}_V$ ; and  $\pi_i$  is inert in  $V$ , since the completion of  $F$  at  $\pi_i$  is  $\mathbb{Q}_2$ —the prime 2 splits—and  $-3 \equiv 5 \pmod{8}$  generates its unramified quadratic extension. Each quotient is therefore the field  $\mathbb{F}_4$ , and as  $\pi_1, \pi_2$  are coprime the Chinese remainder theorem gives a ring isomorphism

$$R/2R \xrightarrow{\sim} R/\pi_1 R \times R/\pi_2 R = \mathbb{F}_4 \times \mathbb{F}_4,$$

under which the two prime reductions are the coordinate lines:  $\pi_1 R/2R = 0 \times \mathbb{F}_4$  and  $\pi_2 R/2R = \mathbb{F}_4 \times 0$ .

Now the unit vectors. A unit vector  $u$  has  $|u|^2 = 1$ , so  $v_{\pi_i}(u) = 0$  at both primes by Lemma 13: its class is invertible in each factor. There are  $3 \cdot 3 = 9$  such classes, and all nine occur. The thirty unit vectors are the  $\omega^k \varphi^j$  with  $0 \leq k < 6$  and  $|j| \leq 2$ , for the same  $\omega = e^{i\pi/3}$  and the half-angle  $\varphi = e^{i\theta_3/2}$  of Section 1; modulo  $2R$ ,  $\omega^3 = -1 \equiv 1$  and  $\varphi^{-1} \equiv \varphi^2$ , so their classes are the nine with  $k, j \in \{0, 1, 2\}$ , and those nine are distinct. So the 9 is a ceiling attained, not a coincidence of a count.

Two lines finish the theorem. The image  $W$  of an index-4 subgroup containing  $2R$  is an  $\mathbb{F}_2$ -subspace of  $\mathbb{F}_4 \times \mathbb{F}_4$  of dimension 2. If it meets no unit class then every element of  $W$  has a zero coordinate, so

$$W = (W \cap (\mathbb{F}_4 \times 0)) \cup (W \cap (0 \times \mathbb{F}_4)),$$

a union of two subgroups—and a group is never the union of two proper subgroups. Hence  $W$  lies inside one coordinate line, and both lines have  $\mathbb{F}_2$ -dimension 2, so  $W$  is that line. Exactly two subgroups survive, and they are the prime reductions:

$$\ker \gamma_1 = \pi_1 R, \quad \ker \gamma_2 = \pi_2 R.$$

The count—two—is the one Dúcz reaches for his own lattice, by enumeration rather than by his sublattice argument, and reached here for  $R$ , which is not his lattice.

Both are geometric, which closes the circle: by Lemma 13,  $v_{\pi_i}(|x|^2) = 2v_{\pi_i}(x)$  for  $x \neq 0$ , so  $x \in \pi_i R$  if and only if  $v_{\pi_i}(|x|^2) \geq 2$ —membership in either kernel is read off the squared length alone, and monochromatic pairs are decided by distance.  $\square$

In coordinates, for checking: take the Hermite basis of  $R$  in the chart (1), rows in the order  $(A, B, C, D)$ ,

$$b_1 = (1, 1, 1, -1), \quad b_2 = (0, 2, 0, -14), \quad b_3 = (0, 0, 2, 2), \quad b_4 = (0, 0, 0, 4),$$

and write a class of  $R/2R \cong \mathbb{F}_2^4$  as the integer whose bit  $i$  is the coefficient of  $b_{i+1}$ . Then

$$\pi_1 R/2R = \{0, 5, 9, 12\}, \quad \pi_2 R/2R = \{0, 3, 4, 7\},$$

the nine unit classes are the nine integers in neither list, and a sweep over all 35 two-dimensional subspaces of  $\mathbb{F}_2^4$ —the enumeration the argument above replaces—returns those two and no others. (The masks depend on the basis and the bit order; both are fixed here so the numbers can be checked.)

Stability is the wrong invariant, and it is worth saying which one fails, because matching the  $\mathcal{O}$ -stable subgroups against Dúcz's two is the natural first move and cannot work. Multiplication by  $\sqrt{33}$  is the *identity* on  $R/2R$ , since  $33 \equiv 1$ , so it constrains nothing; multiplication by  $(1 + \sqrt{33})/2$  leaves eleven of the thirty-five two-dimensional subspaces stable—the two eigenspaces and the nine sums of a line from each—not two. It is avoiding the unit vectors that cuts the field to two, and the ring structure above that names them.

The theorem counts the Klein colourings, and the interface bound of the empty-shell verdicts of Section 7 need the cyclic ones too, so count them here. A homomorphism  $R \rightarrow \mathbb{Z}/4$  missing every unit vector is determined by its values on a basis; writing those as  $\ell + 2s$  over  $\mathbb{F}_2$  makes the condition one affine system per  $\ell$ , a sweep over  $2^4$  rather than  $4^4$ , exactly as in Proposition 6's proof. The answer is small enough to print. In the Hermite basis  $b_1, \dots, b_4$  fixed below, writing a homomorphism as the tuple  $(\varphi(b_1), \dots, \varphi(b_4))$ , exactly twelve of the 256 tuples kill no unit vector:

$$(0, 0, 2, 1), \quad (0, 1, 0, 0), \quad (1, 0, 1, 3), \quad (1, 1, 0, 0), \quad (1, 1, 2, 1), \quad (1, 3, 1, 3)$$

and their negatives. Up to negation—the only non-trivial automorphism of  $\mathbb{Z}/4$ —there are therefore exactly *six*. So  $R$  carries eight homomorphism 4-colourings altogether, two Klein and six cyclic, each counted up to the automorphisms of its own group, and it is that 8 that the interface argument pairs with itself; that computation is `scripts/interface_matchings.py` in the archive.

The colour condition is therefore divisibility in  $\mathcal{O}$ ,

$$\gamma_i(x) = 0 \iff \pi_i \mid x, \tag{4}$$

proved above and confirmed directly, point by point and for both colourings, on a ball of the lattice; the accompanying code reports the size. The number of colourings is two *because 2 splits into exactly two primes*: the subgroups that avoid every unit class are the two prime reductions, and there are two of those because  $33 \equiv 1 \pmod{8}$ . That is a reason, and Dúcz's sublattice proof does not give one. Note also  $\pi_1 R \cap \pi_2 R = \pi_1 \pi_2 R = 2R$ : both colours vanish exactly on  $2R$ .

the archive's drawing draws them.

## 5. PROOF OF THE DIVISIBILITY LAW

Assume  $\Lambda = R + \theta_a(R)$  has rank 8, so  $\Lambda = R \oplus \theta_a(R)$  and a homomorphism  $\varphi : \Lambda \rightarrow \mathcal{A}$  is a pair  $(\varphi_1, \varphi_2)$ , one per summand, and each  $\varphi_i$  must avoid the 30 unit vectors of its own summand.

Take  $\mathcal{A}$  to be the Klein group first. Its exponent is 2, so  $2R \subseteq \ker \varphi_i$  automatically—no geometric hypothesis is needed and none is available, since a homomorphism avoiding the unit vectors need not be geometric: Section 4 exhibits six cyclic ones that are not. What is available is the enumeration inside Theorem 9: a subgroup of index 4 containing  $2R$  and meeting none of the nine unit classes is  $\pi_1 R$  or  $\pi_2 R$ .

The image cannot be smaller than the whole Klein group either. Were it of order 2, then  $1 + \omega + \omega^2 = 2\omega$  would give  $\varphi_i(1) + \varphi_i(\omega) + \varphi_i(\omega^2) = \varphi_i(2\omega) = 0$ , while each of the three terms is the one non-zero element of an order-2 group and their sum is that element again. So each  $\varphi_i$  is reduction modulo  $\pi_1 R$  or  $\pi_2 R$ , and with the six automorphisms of the Klein group that is at most  $2 \cdot 2 \cdot 6 = 24$  candidates, whatever  $a$  is. (Both reductions happen to be geometric, by Theorem 9, but that is a conclusion here and not a hypothesis.)

The cyclic target  $\mathcal{A} = \mathbb{Z}/4$  is disposed of once and for all by Proposition 6, which Step 5 invokes; the six cyclic colourings counted in Section 4 are the ones that survive on  $R$  alone and play no part here. The question is which of them a *cross* unit vector  $u = r + \theta(s)$ , with  $r, s \in R$  both non-zero, destroys. Figure 3 shows the two orbits these fall into.

The five steps deliver, in order: that a cross unit vector has its two parts parallel,  $s = \lambda r$  with  $\lambda$  a rational scalar (Step 1); that  $\lambda = -1$  is the shell, the only case carrying geometric information (Step 2); that a shell point lies in  $2R$  exactly when  $4 \mid a$ , which is where the divisibility enters (Step 3); that every other  $\lambda$  constrains nothing the shell has not already constrained (Step 4); and that the two conclusions follow—a colouring exists when  $4 \nmid a$ , and Proposition 6 forbids one when  $4 \mid a$  with the shell occupied (Step 5).

**Step 1: the two parts are parallel.** Writing  $w = r\bar{s}$ ,

$$|u|^2 = |r|^2 + |s|^2 + \frac{(2a-1)\operatorname{Re} w + \sqrt{4a-1}\operatorname{Im} w}{a}.$$

Separate the coefficients over  $F = \mathbb{Q}(\sqrt{33})$ . By (3) the squared norms  $|r|^2$  and  $|s|^2$  lie in  $F$ , and so does  $\operatorname{Re} w$ ; the chart's imaginary part is  $(C\sqrt{3} + D\sqrt{11})/12$  and  $\sqrt{11} = \sqrt{33}/\sqrt{3}$ , so  $\operatorname{Im} w$  lies in  $\sqrt{3}F$ . Then  $|u|^2 = 1$  reads

$$\alpha + \beta\sqrt{3(4a-1)} = 1, \quad \alpha, \beta \in F,$$

and rank 8 is exactly the statement that  $\sqrt{3(4a-1)} \notin F$ —which by Proposition 4 fails only at  $4a-1 = 3k^2$  or  $11k^2$ . Note this is the  $\mathbb{Q}$ -span and not the field: at  $a = 1/2$  the root  $\sqrt{4a-1} = 1$  lies in  $\mathbb{Q}(\sqrt{3}, \sqrt{11})$  and the rank is 8 regardless. So  $\beta = 0$ , the  $\sqrt{4a-1}$  term vanishes on its own:  $\operatorname{Im} w = 0$ , that is,  $s = \lambda r$  for a real  $\lambda$ .

**Step 2:  $\lambda = -1$  gives the shell.** Then  $u = (1 - \theta)r$ , and  $|1 - \theta| = 1/\sqrt{a}$  forces  $|r|^2 = a$ : the point  $r$  lies on the shell and  $u$  is a shell displacement. Since  $s = -r$  and the colour group has exponent 2,  $\gamma(s) = \gamma(r)$ : the pair of colours is *diagonal*,  $(c, c)$ , and  $\varphi(u) = \psi_1(c) + \psi_2(c)$ , where  $\psi_1, \psi_2$  are the two automorphisms—not the two primes, which keep the letter  $\pi$  throughout. The colour group has exponent 2, so  $\varphi(u) = 0$  exactly when  $\psi_1(c) = \psi_2(c)$ . At  $c = 0$  that holds for *every* candidate, and  $c = 0$  is exactly  $r \in 2R$ , both primes at once. For  $c \neq 0$  it still holds for those candidates whose two automorphisms happen to agree at  $c$ , which is why the conclusion below asks for a relative 3-cycle and not merely for  $c \neq 0$ .

**Step 3:  $r \in 2R \iff 4 \mid a$ .** Here  $S = 144a$  is an integer even when  $a$  is not, because  $A, B, C, D$  are. Then  $v_2(S) = 4 + v_2(a)$ , so  $4 \mid a$ —that is  $v_2(a) \geq 2$ —is exactly  $64 \mid S$ . Recall too that  $|r|^2$  is rational exactly when  $AB + CD = 0$ . Three lemmas settle it.

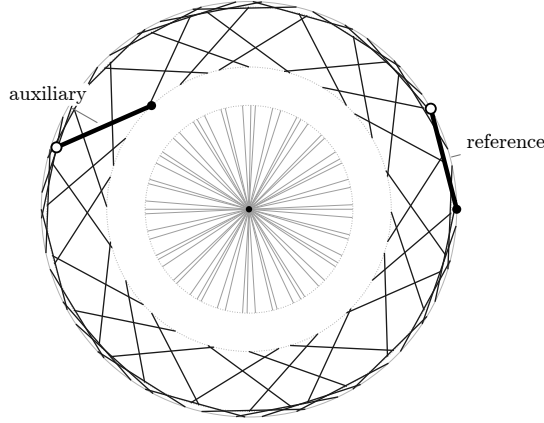


FIGURE 3. The entire interface of  $\theta_4$ : all 126 cross edges a depth-4 ball offers, computed from the coordinates. Sixty run to the shared origin—the inner star—thirty join two points of the shell at radius 2, and thirty-six run crosswise between the dotted radii  $\frac{\sqrt{11}}{2} \mp \frac{\sqrt{3}}{6} = 1.3696$  and  $1.9470$ . Both halves here are the same ball. One edge of each non-trivial orbit is drawn heavy, its end in  $L$  filled and its end in  $\theta_4(L)$  open. Every edge in the figure has length exactly 1.

**Lemma 10.** *For  $(A, B, C, D)$  satisfying (2) with  $AB + CD = 0$  we have  $16 \mid S$ . Equivalently, every rational squared norm of  $R$  lies in  $\frac{1}{8}\mathbb{Z}$ .*

*Proof.* For odd  $x$  write  $x^2 = 1 + 8k_x$ ; then  $x \mapsto k_x \bmod 2$  is the homomorphism  $\varepsilon : (\mathbb{Z}/8)^\times \rightarrow \mathbb{Z}/2$  with  $\varepsilon = 0$  on  $\pm 1$  and 1 on  $\pm 3$ . If all four are odd then  $S \equiv 8(\varepsilon(A) + \varepsilon(B) + \varepsilon(C) + \varepsilon(D)) \pmod{16}$ , and  $AB = -CD$  gives

$$\varepsilon(A) + \varepsilon(B) = \varepsilon(AB) = \varepsilon(-CD) = \varepsilon(C) + \varepsilon(D),$$

so the bracket is even. If all four are even then  $S = 4S'$  with  $S' = A'^2 + 33B'^2 + 3C'^2 + 11D'^2$ , and  $\bmod 4$ ,  $S' \equiv [A'] + [B'] + 3[C'] + 3[D']$  with  $[\cdot]$  the odd indicator; the congruence (ii) makes the number of odd entries 0, 2 or 4: halving (2) gives  $2(A' + B' + C' + D') \equiv 4A' \pmod{4}$ , so  $A' + B' + C' + D'$  is even. Of the six two-element patterns  $\{A', B'\}$  and  $\{C', D'\}$  are excluded by  $A'B' + C'D' = 0$  (one product odd, the other even), while the rest give  $1 + 3 \equiv 0$ .  $\square$

**Lemma 11.** *If  $A, B, C, D$  are all odd and  $AB + CD = 0$  then  $S \equiv 16 \pmod{32}$ . The second hypothesis is not decorative:  $(1, 1, 1, 3)$  gives  $S = 136 \equiv 8$ .*

*Proof.* Refine  $\varepsilon$  one step: for odd  $x$  write  $x^2 = 1 + 8k_x$  as before and set  $\eta(x) = k_x \bmod 4$ , so  $x^2 \equiv 1 + 8\eta(x) \pmod{32}$ . From  $AB = -CD$  we get  $A^2B^2 = C^2D^2$ ,

whose two sides are  $1 + 8(\eta_A + \eta_B)$  and  $1 + 8(\eta_C + \eta_D)$  modulo 32, the cross terms  $64\eta\eta$  vanishing; so  $\eta_A + \eta_B \equiv \eta_C + \eta_D \pmod{4}$ . Expanding  $S$  modulo 32 gives  $16 + 8(\eta_A + \eta_B - \eta_C - \eta_D) \equiv 16$ .  $\square$

**Lemma 12.** *If  $4 \mid X^2 + 33Y^2 + 3Z^2 + 11W^2$  then  $X + Y + Z + W$  is even.*

*Proof.* Modulo 4 a square is the odd indicator, so the form is congruent to  $[X] + [Y] + 3[Z] + 3[W]$ . One odd entry gives 1 or 3; three give  $5 \equiv 1$  or  $7 \equiv 3$ . Neither is 0, so the number of odd entries is even.  $\square$

*Forward.* If  $r = 2w$  with  $w \in R$  then  $a = 4|w|^2$ , and Lemma 10 gives  $|w|^2 = T/9$  for an integer  $T$ , so  $a = 4T/9$  and  $v_2(a) = 2 + v_2(T) \geq 2$ : that is  $4 \mid a$ , with no integrality of  $a$  used. (When  $a$  is an integer the same line reads  $9 \mid T$ .)

*Backward.* If  $4 \mid a$  then  $64 \mid S$ , so  $32 \mid S$ , which Lemma 11 forbids for all-odd coordinates; by (2) they are all even, and with  $A = 2A'$  and so on,  $S = 4S'$  with  $16 \mid S'$  and  $A'B' + C'D' = 0$ . The halves satisfy (i): Lemma 12 gives an even number of odd entries among them, and two is impossible. Suppose exactly two of  $A', B', C', D'$  are odd. The pairs  $\{A', B'\}$  and  $\{C', D'\}$  are out at once: either would make one side of  $A'B' = -C'D'$  odd and the other even. That leaves the four *mixed* pairs, one entry from  $\{A', B'\}$  and one from  $\{C', D'\}$ , and they all behave the same way, for a reason visible in the coefficients of  $S' = A'^2 + 33B'^2 + 3C'^2 + 11D'^2$ :

$$1 + 3 = 4, \quad 1 + 11 = 12, \quad 33 + 3 = 36, \quad 33 + 11 = 44,$$

every one of them  $\equiv 4 \pmod{8}$ . An odd square is  $\equiv 1 \pmod{8}$ , so the two odd terms of  $S'$  contribute 4 modulo 8. For the two even ones,  $A'B' = -C'D'$  forces them to have the same 2-adic valuation  $k \geq 1$ —the odd entry on each side contributes nothing—so together they contribute  $2^{2k}(\alpha b^2 + \beta d^2)$  with  $b, d$  odd and  $\{\alpha, \beta\}$  the complementary pair of coefficients, which is again  $\equiv 4 \pmod{8}$ . At  $k = 1$  that is  $4 \cdot (8j + 4) \equiv 0 \pmod{16}$ , and at  $k \geq 2$  the factor  $2^{2k}$  already kills it. Either way  $S' \equiv 4 \pmod{8}$ , so  $S'$  is 4 or 12 modulo 16 and never 0, against  $16 \mid S'$ . They satisfy (ii): when all odd, reducing  $A'B' = -C'D' \pmod{4}$  with every entry in  $\{1, 3\}$  leaves an odd number congruent to 3, which is exactly (ii); when all even it reduces to  $A'' + B'' + C'' + D''$  being even, and  $16 \mid S'$  gives  $4 \mid S''$ , so Lemma 12 supplies it. Hence  $r \in 2R$ .  $\square$

**Step 4: every other  $\lambda$  imposes nothing new.** Both  $r$  and  $s$  lie in  $V = \mathbb{Q}(\sqrt{33}, i\sqrt{3})$ , which Proposition 4 says is a field, so  $\lambda = s/r$  lies in  $V$  too; being real it lies in  $V \cap \mathbb{R} = \mathbb{Q}(\sqrt{33})$ , the only real quadratic subfield. Write  $\lambda = \alpha/\beta$  with  $\alpha, \beta$  coprime in  $\mathcal{O}$ —which is possible because  $\mathcal{O}$  is a principal ideal domain,  $\mathbb{Q}(\sqrt{33})$  having class number one—and put  $v = r/\beta$ . It lies in  $R$ : Bézout gives  $x\alpha + y\beta = 1$ , and then  $v = xs + yr \in R$  since  $R$  is an  $\mathcal{O}$ -module, with  $\beta v = r$  and  $\alpha v = s$ . Substituting  $r = \beta v$ ,  $s = \alpha v$  into  $|u|^2 = 1$  and clearing denominators,

$$|v|^2 [a(\alpha - \beta)^2 + (4a - 1)\alpha\beta] = a. \quad (5)$$

**Lemma 13.** *For non-zero  $v \in R$  and  $\pi$  a prime of  $\mathcal{O}$  above 2,  $v_\pi(|v|^2) = 2v_\pi(v)$ , where  $v_\pi(v)$  is the largest  $k$  with  $v \in \pi^k R$ .*

*Proof.* First,  $9|x|^2 \in \mathcal{O}$  for every  $x \in R$ —Lemma 10 is its rational case—because  $144|\cdot|^2$  is a quadratic form with values in  $\mathcal{O}$ , and both its values and its polarisations on the 30 unit vectors are divisible by 16 there: 30 checks and  $\binom{30}{2} = 435$ . So  $v_\pi(144|x|^2) \geq 4 = v_\pi(144)$  throughout  $R$ .

Now write  $v = \pi^k v_1$  with  $\pi \nmid v_1$ . Since  $\pi$  is real,  $|v|^2 = \pi^{2k} |v_1|^2$ , so it is enough that  $|v_1|^2$  be a  $\pi$ -unit, that is that  $v_\pi(144|v_1|^2) = 4$  exactly. Replacing  $v_1$  by  $v_1 + \pi w$  changes  $144|v_1|^2$  by  $144\pi(v_1 \bar{w} + \bar{v}_1 w) + 144\pi^2 |w|^2$ , of valuation at least 5 by the previous paragraph, so a valuation of 4 depends only on the class of  $v_1$  in  $R/\pi R \cong \mathbb{F}_2^2$ . Its three non-zero classes are represented by the six points of least norm,  $|v|^2 = 1/9$ —namely  $\pm \frac{1}{3}$  and  $\pm \frac{1}{6} \pm \frac{i\sqrt{3}}{6}$ —and there  $144|v|^2 = 16 = (\pi_1 \pi_2)^4$ , of valuation exactly 4.  $\square$

**Lemma 14.** *Let  $\pi$  be a prime of  $\mathcal{O}$  above 2, and let the denominator of  $a$  be odd—which holds automatically in the only case Corollary 15 uses, namely  $a$  even, since then the denominator of  $a$  in lowest terms is odd; more generally Theorem 18 forces it whenever the shell is non-empty, since it puts the denominator in  $\{1, 3, 9\}$ . If  $\pi$  divides none of  $v, \alpha, \beta$ , then  $a$  is odd.*

*Proof.* The residue field  $\mathcal{O}/\pi$  is  $\mathbb{F}_2$ , so  $\alpha \equiv \beta \equiv 1$  there, hence  $\alpha - \beta \equiv 0$  and  $\alpha\beta \equiv 1$ , and the bracket in (5) is  $\equiv 4a - 1 \equiv 1 \pmod{\pi}$ : writing  $a = n/b$  with  $b$  odd makes  $4n - b$  odd, so  $v_\pi(4a - 1) = 0$ . (With  $4 \mid b$  that fails, which is why the hypothesis is stated.) So  $\pi$  does not divide it, and (5) gives  $v_\pi(|v|^2) = v_\pi(a) = v_2(a)$ . With  $\pi \nmid v$ , Lemma 13 makes the left side 0.  $\square$

**Corollary 15.** *If  $4 \nmid a$  then neither prime divides  $v$ . If moreover  $a$  is even then  $\pi_i$  divides exactly one of  $r = \beta v$  and  $s = \alpha v$ , so by (4) exactly one of  $\gamma_i(r), \gamma_i(s)$  vanishes and  $\varphi(u) = 0 + c$  with  $c \neq 0$ : the vector imposes no condition on a candidate colouring both summands the same way—which is all Step 6 asks for—for every  $\lambda$ , rational or not.*

*Proof.* Write  $a = n/b$  in lowest terms and clear  $b$  in (5):

$$|v|^2 [n(\alpha - \beta)^2 + (4n - b)\alpha\beta] = n,$$

a bracket that lies in  $\mathcal{O}$  whatever the parities, where the bracket of (5) need not. By Lemma 13 the factor  $|v|^2$  has valuation  $2v_\pi(v)$ , so the left side has valuation  $2v_\pi(v) + v_\pi(\text{bracket})$ ; the bracket lies in  $\mathcal{O}$ , so that second term is at least 0, and comparing with the right gives  $2v_\pi(v) \leq v_2(n)$ —an inequality, which is all the proof uses. If  $b$  is odd then  $v_2(n) = v_2(a) \leq 1$  because  $4 \nmid a$ ; if  $b$  is even then  $n$  is odd and  $v_2(n) = 0$ . Either way  $v_\pi(v) = 0$ .

Now let  $a$  be even. Lemma 14 puts  $\pi$  into one of  $v, \alpha, \beta$ ; the first is excluded, and  $\alpha, \beta$  are coprime, so exactly one of them is divisible—say  $\pi \mid \beta$ . Then  $\pi \mid r$ , while  $\alpha \equiv 1$  and  $v \not\equiv 0 \pmod{\pi}$  leave  $\gamma_i(s) = \gamma_i(v) \neq 0$ .  $\square$

The first half of the corollary settles the whole of Steps 4 and 5 by itself, for every  $\lambda$  at once, and it is worth saying so before the case analysis that found it. Fix a prime  $\pi_i$ . Since  $\mathcal{O}/\pi_i = \mathbb{F}_2$ , an  $\alpha$  the prime misses satisfies  $\alpha \equiv 1$ , so  $\alpha v \equiv v$  modulo  $\pi_i R$ ; hence  $\gamma_i(s) = \gamma_i(\alpha v)$  is 0 if  $\pi_i \mid \alpha$  and  $\gamma_i(v)$  if not, and likewise  $\gamma_i(r) = \gamma_i(\beta v)$ . As  $\alpha, \beta$  are coprime the prime misses at least one of them, leaving three cases and only three, for  $4 \nmid a$ :

- $\pi_i \mid \beta$ . Then  $\gamma_i(r) = 0$  while  $\gamma_i(s) = \gamma_i(v) \neq 0$ , so  $\varphi(u) \neq 0$  for every candidate: the vector imposes nothing at all.
- $\pi_i \mid \alpha$ . The same with  $r$  and  $s$  exchanged.
- Neither. Then  $\gamma_i(r) = \gamma_i(s) = \gamma_i(v)$  and the pair is the diagonal  $(c, c)$  of Step 2, with  $c \neq 0$ .

So a cross vector with  $\lambda \neq -1$  imposes nothing the shell displacements do not, whatever  $\lambda$  is—rational or not,  $a$  odd or even. That disposes of every  $\lambda$ , at odd and even  $a$  alike, and the proof can go straight to the conclusion. The finer analysis the trichotomy replaced—which  $\lambda$  actually occur at odd  $a$ —and the sweep that checks both against every rank-8 spindle to  $a = 80$  are in the repository, as `docs/odd-case.tex` and `scripts/lambda_sweep.py`; nothing here depends on either.

**Step 5: conclusion.**

**Lemma 16.** *For a shell point  $r$ , either both primes divide  $r$ —which happens exactly when  $4 \mid a$ —or neither does.*

*Proof.* Both divide if and only if  $r \in \pi_1 R \cap \pi_2 R = 2R$ , which by Step 3 is  $4 \mid a$ . Suppose exactly one does, say  $r = \pi_1 w$ . Then  $|w|^2 = a/\pi_1^2 = a(29 - 5\sqrt{33})/8$ , so in the chart  $w$  has

$$S = 522a, \quad 2(AB + CD) = -90a.$$

A non-empty shell puts  $a$  in  $\frac{1}{9}\mathbb{Z}$  by Lemma 10, and 9 divides both 522 and 90, so these are integers; it also makes  $v_2(a) \geq 0$ , so the two cases below exhaust  $4 \nmid a$ . If  $a$  is odd then  $S \equiv 2 \pmod{4}$ , and  $S \equiv [A] + [B] + 3[C] + 3[D]$  takes the value 2 only on the patterns oooo and eooo, both of which violate (i). If  $v_2(a) = 1$ , write  $a = 2a'$  with  $a'$  odd; then  $S \equiv 4 \pmod{8}$ , which all-odd coordinates cannot give since they force  $S \equiv 0 \pmod{8}$ , so all coordinates are even and  $AB + CD = 4(A'B' + C'D')$ , while  $AB + CD = -45a = -90a'$  is  $2 \pmod{4}$ . The case  $r = \pi_2 w$  is the image of this one under the Galois conjugation  $\sigma$  of  $F$ , which swaps  $\pi_1$  and  $\pi_2$  and changes only the sign of  $90a$ ; every congruence above is a statement modulo 4 or 8 about  $S = 522a$  and  $AB + CD = \mp 45a$ , and none of them sees that sign.  $\square$

If  $4 \mid a$  and the shell is non-empty—this is the only place the second hypothesis is used, and the only half of the theorem that needs it—Step 3 gives a shell point  $r = 2w$ , and then  $(1 - \theta_a)w$  lies in  $\Lambda_a$  with length  $\frac{1}{2}$ , so Proposition 6 rules out both groups of order 4 at once. Concretely, the shell displacement  $u = (1 - \theta_a)r$  has both colours zero, so  $\varphi(u) = 0$  for all 24 Klein candidates and none survives. If  $4 \nmid a$ , Lemma 16 says no shell point is divisible by either prime, so both colourings give it a non-zero colour: the shell displacements impose only diagonal pairs  $(c, c)$  with  $c \neq 0$ , uniformly over the shell and for either choice of colouring. Fix one of the two and use it on both summands, which is what makes the pairs diagonal at all. By Step 4 nothing else constrains, and two automorphisms differing at every non-zero point satisfy  $\psi_1(c) \neq \psi_2(c)$ —the relative permutation must be a 3-cycle, and two of the six are—so at least four of the twenty-four candidates survive, and a colouring exists. This proves Theorem 1.  $\square$

## 6. THE DISTANCE SPECTRUM, AND THE CANDIDATES IT LEAVES

The law leaves  $4 \mid a$  with a non-empty shell as the candidates (Figure 2). Which  $a$  have one is decided by the form (3) and not by a ball, and has a closed answer, because  $R$  is  $\frac{1}{3}\mathcal{O}_V$  and occupancy is a norm question.

Call a prime  $p \neq 3, 11$  *obstructed* when it splits in  $\mathbb{Q}(\sqrt{33})$  and stays inert in  $\mathbb{Q}(\sqrt{-3})$ . For odd  $p$  that is the pair of symbols

$$\left(\frac{33}{p}\right) = 1 \quad \text{and} \quad \left(\frac{-3}{p}\right) = -1,$$

and neither need be computed: since  $33 \equiv 1 \pmod{4}$ , quadratic reciprocity gives  $\left(\frac{33}{p}\right) = \left(\frac{p}{33}\right)$ , and  $\left(\frac{-3}{p}\right) = 1$  exactly when  $p \equiv 1 \pmod{3}$ . Both are therefore fixed by  $p$  modulo 33, and running the 20 classes coprime to 33 leaves five:

$$p \text{ is obstructed} \iff p \equiv 2, 8, 17, 29 \text{ or } 32 \pmod{33}.$$

The prime 2 is obstructed too, straight from the definition— $33 \equiv 1 \pmod{8}$  splits it,  $-3 \equiv 5 \pmod{8}$  keeps it inert—and it lands in the class 2  $\pmod{33}$  with the others, so the residue form covers it as well and nothing below makes it a separate case. Below 200 the obstructed primes are 2, 17, 29, 41, 83, 101, 107, 131, 149, 167, 173 and 197; the residue form is checked against the symbols for every odd prime below 20000.

**Remark 17** (The five classes are a genus). The five classes are not an artefact of running twenty. Since  $h(F) = 1$  and the fundamental unit has norm +1, the narrow class number of  $F = \mathbb{Q}(\sqrt{33})$  is  $h^+(F) = 2$ , so  $F$  has exactly one quadratic extension unramified at every finite prime. It can be named without any computation:  $\text{disc}(F) = 33 = (-3) \cdot (-11)$  is a product of two prime discriminants, so the narrow genus field of  $F$  is  $\mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ , and since  $\sqrt{-3}\sqrt{-11} = -\sqrt{33}$  that field is  $V$ . So

$V$  is the narrow Hilbert class field of  $\mathbb{Q}(\sqrt{33})$ .

This gives  $\mathfrak{d}_{V/F} = (1)$  a second time, by a route independent of the discriminant comparison in Proposition 8; the two agree, and either may be taken as the source.

What it buys is that the local behaviour used above and below is class field theory rather than case work: a prime of  $F$  splits in  $V$  exactly when it has a *totally positive* generator. The three cases the paper needs fall out at sight. The primes above 2 are generated by  $(\pm 5 \pm \sqrt{33})/2$ , of norm  $-2$ , whose embeddings  $+5.372$  and  $-0.372$  have opposite signs; no unit can repair that, since the units are  $\pm \varepsilon^k$  with  $N(\varepsilon) = +1$ —so  $\pi_1$  and  $\pi_2$  are inert, which is what Section 4 found by hand.  $\mathfrak{p}_3 = (6 + \sqrt{33})$  embeds as  $+11.745$  and  $+0.255$ , is totally positive, and splits.  $\mathfrak{p}_{11} = (11 + 2\sqrt{33})$  embeds as  $+22.489$  and  $-0.489$  and does not.

Over  $\mathbb{Q}$  the same statement is genus theory of discriminant 33. The genus characters are  $\left(\frac{\cdot}{3}\right)$  and  $\left(\frac{\cdot}{11}\right)$ , there are two classes of binary quadratic forms of that discriminant and one in each genus, and  $p$  is obstructed exactly when both characters are  $-1$ :  $p$  lies in the non-principal genus of discriminant 33. That is the same five classes, and it explains why five. Half of the twenty classes coprime to 33 fail the first character, and half of those fail the second.

**Theorem 18.** *Let  $a = n/b$  be a positive rational in lowest terms. Then  $R$  has a point at squared radius  $a$  if and only if  $b \in \{1, 3, 9\}$  and  $v_p(n)$  is even at every obstructed  $p$ .*

*Proof.* By Proposition 8,  $a$  is a squared radius of  $R$  if and only if  $9a = N_{V/F}(w)$  for some  $w \in \mathcal{O}_V$ . Write  $\varepsilon = 23 - 4\sqrt{33}$ , a generator of the totally positive units of  $F$ —the unit group is  $\langle -1 \rangle \times \langle \varepsilon \rangle$  and  $-1$  is negative at both real places, so those units are exactly the powers of  $\varepsilon$ . Three further facts are used, and it is worth saying where each comes from.

$V/F$  is cyclic of degree 2 and unramified at every finite prime: that is the by-product of Proposition 8's discriminant comparison, not a new assumption.

$h(F) = 1$ , in two lines. The Minkowski bound for  $F = \mathbb{Q}(\sqrt{33})$  is  $\frac{1}{2}\sqrt{33} < 2.88$ , so every ideal class contains an integral ideal of norm at most 2, that is the unit

ideal or a prime above 2. Both of those are principal, since  $2 = -\pi_1\pi_2$  with  $\pi_{1,2} = (5 \pm \sqrt{33})/2$  in  $\mathcal{O}_F$ . So the class group is trivial.

$\varepsilon = 23 - 4\sqrt{33}$  generates the totally positive units of  $F$ —it is the inverse of the fundamental unit  $23 + 4\sqrt{33} \approx 45.98$  under the usual convention, and generates the same group—and it is totally positive:  $23^2 - 33 \cdot 4^2 = 1$ , and both conjugates  $23 \mp 4\sqrt{33}$  are positive. Since  $-1$  is negative at both real places, the totally positive units are exactly the powers of  $\varepsilon$ .

$h(V) = 1$ , by Minkowski again, and the computation is small enough to give in full. With  $n = 4$ ,  $r_2 = 2$  and  $|\text{disc}(V)| = 1089$  the bound is

$$\left(\frac{4}{\pi}\right)^2 \frac{4!}{4^4} \sqrt{1089} = \frac{16}{\pi^2} \cdot \frac{3}{32} \cdot 33 < 5.02,$$

so every ideal class contains an integral ideal of norm at most 5. Which primes of  $\mathcal{O}_V$  have norm that small? Above 2: each  $\pi_i$  is inert in  $V$ , so the prime above it has norm  $N(\pi_i)^2 = 4$ , and there is no prime of norm 2. Above 3: the prime  $\mathfrak{p}_3$  of  $F$  ramifies, and splits in  $V$  into two primes of norm 3. Above 5:  $33 \equiv 3 \pmod{5}$  is a non-residue, so 5 is inert in  $F$  and every prime above it has norm at least 25. Hence the only integral ideals of norm at most 5 are the unit ideal, the two primes of norm 3 and the two of norm 4, and it is enough to generate those four.

The ones above 2 are  $\pi_1\mathcal{O}_V$  and  $\pi_2\mathcal{O}_V$ , principal already in  $\mathcal{O}_F$ . For those above 3, take

$$w = \frac{1}{2}(1 + i(2\sqrt{3} + \sqrt{11})) \in \mathcal{O}_V,$$

which lies in  $3R$  because  $w/3$  is the chart point  $(2, 0, 4, 2)$  of (1)—the point  $(6, 0, 12, 6)$  verifies only  $w \in R$ —and has

$$N_{V/F}(w) = |w|^2 = \frac{1}{4}(1 + (2\sqrt{3} + \sqrt{11})^2) = \frac{1}{4}(24 + 4\sqrt{33}) = 6 + \sqrt{33},$$

so  $N_{V/\mathbb{Q}}(w) = 36 - 33 = 3$  and  $(w)$  is one of the two; its conjugate under  $\sqrt{33} \mapsto -\sqrt{33}$  is the other. So the class group is trivial.

(PARI agrees, and `scripts/moser_is_ov.py` runs the check, but the argument above does not need it.)

Which primes of  $F$  are inert in  $V$ ? Exactly those above an obstructed  $p$ , together with the one above 11. That is the content of the two symbols:  $p$  splits in  $F$  and stays inert in  $\mathbb{Q}(\sqrt{-3})$  precisely when the prime above it does not split in  $V$ .

*Necessity.* Let  $9a = N(w)$  with  $w \in \mathcal{O}_V$ . Then  $9a$  lies in  $\mathcal{O}_F \cap \mathbb{Q} = \mathbb{Z}$ , so  $b \mid 9$ . At a prime  $P$  of  $F$  inert in  $V$ ,  $v_P(N(w)) = 2v_{P'}(w)$  is even. For  $P$  above an obstructed  $p$  we have  $v_P = v_p$  on rationals, so  $v_p(9a)$  is even; as no obstructed  $p$  equals 3,  $v_p(a)$  is even too. (Above 11 the prime ramifies in  $F$ , so  $v_P = 2v_{11}$  is even for every rational and the condition is empty.)

*Sufficiency.* Suppose  $b \mid 9$  and  $v_p(a)$  is even at every obstructed  $p$ . Then  $9a$  is a positive rational, hence totally positive in  $F$ , and has even valuation at every prime of  $F$  inert in  $V$ . At the two infinite places  $V$  is totally imaginary, so the local norms there are the positive reals, which  $9a$  meets; at a finite prime that splits there is no condition, and at an inert one the condition is even valuation, so  $9a$  is a local norm everywhere and Hasse's norm theorem for the cyclic extension  $V/F$  gives  $9a = N(x)$  for some  $x \in V^\times$ . We record that because it is the natural first move, but the argument does not use it: what is wanted is an algebraic *integer*, and the route to one goes through the ideal directly, with no appeal to Hasse at any point.

Build the ideal, then. Even valuation at the inert primes is exactly what lets  $(9a)\mathcal{O}_F$  be written as  $N_{V/F}(A)$  for an ideal  $A$  of  $\mathcal{O}_V$ ; since  $h(V) = 1$ ,  $A = (w)$ , and  $N(w)$  generates  $(9a)$ , so  $N(w) = 9a u$  for a unit  $u$  of  $\mathcal{O}_F$ . Both  $9a$  and  $N(w) = |w|^2$  are totally positive, so  $u$  is. The totally positive units of  $F$  are the powers of  $\varepsilon$ —the unit group is  $\langle -1 \rangle \times \langle \varepsilon \rangle$  and  $-1$  is negative at both places—and  $\varepsilon = N_{V/F}(\eta)$  for

$$\eta = i(2\sqrt{3} - \sqrt{11}) \in \mathcal{O}_V,$$

which is purely imaginary, so  $\bar{\eta} = -\eta$  and

$$N_{V/F}(\eta) = \eta\bar{\eta} = (2\sqrt{3} - \sqrt{11})^2 = 12 + 11 - 4\sqrt{33} = 23 - 4\sqrt{33} = \varepsilon,$$

while  $N_{V/\mathbb{Q}}(\eta) = 23^2 - 16 \cdot 33 = 1$ , so  $\eta$  is a unit. It lies in  $\mathcal{O}_V = 3R$  because  $\eta/3 = i(2\sqrt{3} - \sqrt{11})/3$  is the chart point  $(0, 0, 8, -4)$ . Replacing  $w$  by  $w\eta^{-k}$  with  $u = \varepsilon^k$  gives  $N(w) = 9a$  exactly.  $\square$

The ingredients are all small and all checkable:  $h(F) = h(V) = 1$ ,  $\text{disc}(V) = 1089$ , the signature  $(0, 2)$ , and  $N_{V/F}(\eta) = \varepsilon$ , which the accompanying code verifies along with  $3R$  being the maximal order. Where the parity comes from is set out below, after the density.

The denominator contributes nothing beyond being admissible, which is consistent: 3 is not obstructed, and the reason is local rather than a matter of ramification—11 is a square in  $\mathbb{Q}_3$ , so  $\sqrt{-3}$  already lies in  $\mathbb{Q}_3(\sqrt{33})$  and the extension splits there, leaving no condition. At 11 the extension is inert, but 11 ramifies in  $\mathbb{Q}(\sqrt{33})$ , so a rational has even valuation at the prime above it and the condition is empty again. The restriction  $b \in \{1, 3, 9\}$  is Lemma 10's corollary from Section 5, and with it the criterion covers *every* rational radius, not merely the integers. The three rational spindles found below are all  $n = 64$  or 256, powers of two with even exponent, and all three have non-empty shells.

Put beside Theorem 1 this describes the occupied-shell candidates off the factorisation of  $a$ , with no ball and no search. It does not say which candidates admit a 5-chromatic join: Section 7 kills eleven of the sixteen below 100;  $a = 64$  is the one it cannot reach, and Lemma 20 settles that one by construction instead.

**Corollary 19.** *Let  $a \geq 1/2$ , normalised as in the introduction, and write  $a = n/b$  in lowest terms. Then  $\theta_a$  is a candidate—the law admits it and its shell is non-empty—if and only if*

$$b \in \{1, 3, 9\}, \quad v_p(n) \text{ is even at every obstructed } p, \quad v_2(n) \geq 2.$$

Only the last clause is new: 2 being obstructed, the middle clause already asks for  $v_2$  even, and  $4 \mid a$  adds  $v_2 \geq 2$ . That single line accounts for the nine empty multiples of 4 below 100 without any computation—eight have  $v_2$  odd, and the ninth is  $68 = 2^2 \cdot 17$ , where the obstructed prime 17 appears once—and it was checked against the form for every integer  $a \leq 400$  without exception.

It also settles how common candidates are. Writing  $\mathcal{C}$  for the integer candidates, each condition restricts at one prime and distinct primes are independent, so for any finite set  $P$  of obstructed primes containing 2,

$$\bar{d}(\mathcal{C}) \leq \frac{1}{6} \prod_{\substack{p \in P \\ p \neq 2}} \frac{p}{p+1},$$

the  $\frac{1}{6}$  being 2's own share. Five of the twenty classes mod 33 are obstructed, so those primes have Dirichlet density  $\frac{1}{4}$ ,  $\sum 1/p$  diverges over them, and the product tends to 0: the integer candidates have natural density zero. The measured densities to  $N = 10^6$ , the Mertens estimate for the rate, and the cautions about comparing a finite count with an asymptotic bound are in the archive, with the script that produces them.

| rotation         | cos     | field                                          | depth | radius | vertices ( $\chi \geq 5$ ) |
|------------------|---------|------------------------------------------------|-------|--------|----------------------------|
| $\theta_4$       | 7/8     | $\mathbb{Q}(\sqrt{3}, \sqrt{5}, \sqrt{11})$    | —     | —      | 509, minimised             |
| $\theta_{16}$    | 31/32   | $\mathbb{Q}(\sqrt{3}, \sqrt{7}, \sqrt{11})$    | 4     | 4.0000 | 29113                      |
| $\theta_{28}$    | 55/56   | $\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{37})$   | 6     | 5.3113 | 156577                     |
| $\theta_{36}$    | 71/72   | $\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{13})$   | 6     | 6.0000 | 175825                     |
| $\theta_{64/3}$  | 125/128 | $\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{23})$   | 7     | 4.6383 | 187225                     |
| $\theta_{64/9}$  | 119/128 | $\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{247})$  | 3     | 2.6817 | 5809                       |
| $\theta_{256/9}$ | 503/512 | $\mathbb{Q}(\sqrt{3}, \sqrt{11}, \sqrt{1015})$ | 6     | 5.3506 | 158257                     |

The first is Parts' record and was known; the four of [9] are the other previously known rotations, and  $\theta_4$  is common to both lists. Of the six new ones, the three rational spindles follow from Parts' published monochromatic-pair gadgets by the chaining below, and  $\theta_{16}$ ,  $\theta_{28}$ ,  $\theta_{36}$  are the three for which we found no prior construction. Every row carries a DRAT certificate that `drat-trim` checks, shipped in the archive at the commit pinned in *Code and data* below, so the verdicts here stand on the archive and not on any unpublished work: eight proofs, all reporting `s VERIFIED`, re-checkable in about ten minutes by `scripts/check_drat.sh`.

The two lemmas the introduction leans on are short enough to keep.

**Lemma 20** (chaining a monochromatic pair). *Let  $M \subset R$  be finite with two marked points 0 and  $t \neq 0$  such that every proper 4-colouring of  $M$  gives them the same colour, and let  $n \geq 1$  be such that  $a = n^2|t|^2$  is rational and exceeds  $\frac{1}{4}$ , so that  $\theta_a$  is one of the rotations of Section 1. Put  $M_n = \bigcup_{j=0}^{n-1} (M + jt)$ . Then  $M_n \subset R$ , every proper 4-colouring of  $M_n$  gives 0 and  $nt$  the same colour, and  $|M_n| \leq n(|M| - 1) + 1$ . Consequently, writing  $a = n^2|t|^2$ ,*

$$M_n \cup \theta_a(M_n)$$

*is not 4-colourable, on at most  $2n(|M| - 1) + 1$  vertices.*

*Proof.*  $R$  is a group and  $t \in R$ , so every translate  $M + jt$  lies in  $R$  and so does the union;  $a > \frac{1}{4}$  makes  $\sqrt{4a - 1}$  real, so  $\theta_a$  exists, and  $a \in \mathbb{Q}$  makes it a spindle. In a proper 4-colouring the copy  $M + jt$  forces  $jt$  and  $(j + 1)t$  to agree, so 0 and  $nt$  agree; the copies overlap in at least the shared marked point, which gives the count. The same argument inside  $\theta_a(M_n)$  forces  $\theta_a(0) = 0$  and  $\theta_a(nt)$  to agree, and the two copies share the origin, so  $nt$  and  $\theta_a(nt)$  both carry the colour of the origin. But  $|nt|^2 = a$  puts  $nt$  on the shell, so  $|nt - \theta_a(nt)| = 1$  and they are adjacent.  $\square$

**Lemma 21.** *Let  $G = L \cup \theta(S)$  be a finite non-4-colourable join sharing the origin. Then some induced subgraph of  $G$  containing the origin is a join of the same form and has chromatic number exactly 5.*

*Proof.* Among the induced subgraphs of  $G$  that contain the origin and are not 4-colourable—a non-empty family, since  $G$  is one—take one,  $H$ , with fewest vertices. Then  $\chi(H) \geq 5$ . For the other direction pick any  $v \neq 0$  of  $H$ ; by minimality  $H - v$

has a proper 4-colouring, and giving  $v$  a fifth colour extends it to  $H$ , so  $\chi(H) \leq 5$ . And  $H$  is a join of the same form: putting

$$L_H = L \cap V(H), \quad S_H = \theta^{-1}(V(H) \cap \theta(S)),$$

we have  $V(H) = L_H \cup \theta(S_H)$  with  $L_H \subseteq L$  and  $S_H \subseteq S$ , and  $0 \in V(H)$  by construction.  $\square$

Lemma 20 applied to Parts' 367-vertex gadget, whose two terminals lie  $8/3$  apart, gives a join at  $a = 64n^2/9$  for every  $n \geq 1$ ; its premise is the gadget's monochromatic-pair property, which is itself one of the eight certificates.

## 7. A STRONGER FILTER, AND WHAT SURVIVES IT

The criterion of Section 2 uses group homomorphisms, which are exactly the *linear* proper colourings of  $R/mR$ . Every proper colouring of that finite quotient pulls back to the whole lattice, linear or not, so using all of them is strictly stronger and still needs no ball. A periodic colouring sees a cross pair only through its residues, so all it can see is which *residue pairs* occur, and  $(\alpha, \beta)$  occurs exactly when the coset  $(\alpha - \theta\beta) + m\Lambda$  contains a unit vector of the joint lattice.

One degeneracy must be excluded: if  $(0, 0)$  occurs—equivalently if  $\Lambda$  holds a vector of length  $1/m$ —both halves give the centre's coset the same colour and the test cannot kill. At  $m = 2$  that is precisely the half-unit criterion of Section 3. The rule followed is the obvious one: take the smallest modulus at which  $(0, 0)$  does not occur, and report it; a rotation degenerate at every modulus tried is set aside as undecided rather than as a survivor.

Applied to the integer spindles to  $a = 100$ , it kills  $\theta_a$  for  $a = 12, 20, 44, 48, 52, 60, 76, 80, 84, 92$  and  $100$ —nine at  $m = 4$  over  $4^4 = 256$  residue classes, and  $a = 48, 80$ , degenerate at  $m = 4$  and  $m = 6$ , at  $m = 8$  over 4096. Each kill ships as a witness naming its modulus, with both colourings and every occurring residue pair, re-checked to be proper, to agree at the shared origin and to disagree on every occurring pair. That is the whole infinite union, not a sample of it.

That leaves  $\theta_4, \theta_{16}, \theta_{28}, \theta_{36}$ —and  $\theta_{64}$ , where  $(0, 0)$  occurs at every modulus tried, so the filter never speaks. It never could have: Lemma 20 exhibits a join at  $a = 64$ , and no colouring argument excludes what exists. The degeneracy is forced rather than unlucky, since  $R \cap \mathbb{Q} = \frac{1}{3}\mathbb{Z}$  makes  $(8/m)(1 - \theta_{64})$  a vector of length  $1/m$  in  $\Lambda_{64}$  whenever  $8/m \in \frac{1}{3}\mathbb{Z}$ , which holds for every  $m \mid 24$ ; the four moduli tried all divide 24.

The empty-shell multiples of 4 lie outside this tally, since the law is silent there, and are settled separately. There are nine below 100. Six die:  $a = 8, 32, 40, 56, 68$  and  $96$  each carry a homomorphism 4-colouring—exactly 96 apiece, 24 Klein and 72 cyclic, counted as in Section 4—so no subset of the joint lattice is 5-chromatic there. Three survive:  $a = 24, 72$  and  $88$  admit none, and the reason is Proposition 6 rather than anything in this section—each carries exactly twelve vectors of length  $\frac{1}{2}$ , one of  $\Lambda_{24}$ 's written out in Section 3, so no homomorphism onto a group of order 4 can avoid the unit vectors whatever the shell does. An earlier version of the accompanying code declined to run the periodic filter at an empty shell, on the ground that the question did not arise. It does arise, and these three are what the refusal was hiding. Each of the nine ships a witness that `scripts/check_emptyshell.py` validates from its JSON alone.

## 8. WHAT IS NOT CLAIMED

The criterion is a one-sided kill. A survivor is a candidate; only an explicit 5-chromatic union makes it a construction, which is why Section 6 exhibits graphs rather than citing survivals.

The law is about *this* lattice, and the same criterion run on Haugland's lattice  $L_1$ —the one generated by the 42nd roots of unity—is the evidence:  $\theta_2$  dies on the Moser lattice and survives there, as does  $\theta_7$ . Those computations are `scripts/hept_rank.py`, `hept_classify.py` and `hept_halfunit.py` in the archive. The divisibility law does not transfer, and no attempt is made to transfer it.

The rank-8 hypothesis excludes rotations that map  $R$ 's span into itself— $\theta_3$  is  $\omega_3$ ,  $\theta_1$  is  $\omega_1$ —where the cross decomposition does not arise. Proposition 4 says exactly which those are,  $4a - 1 = 3k^2$  or  $11k^2$ , so the hypothesis is recognisable at sight rather than checkable per rotation, and it is proved rather than assumed.

Proposition 6 is unconditional for both groups of order 4. An earlier version of it was not, and the remark in Section 3 records why: the three exceptions it reported were a bug in the sweep, not a feature of the lattice.

Theorem 18 is proved, through Proposition 8:  $R$  is  $\frac{1}{3}\mathcal{O}_V$ , so occupancy is a question about relative norms, which the arithmetic of  $\mathcal{O}_V$  answers—by  $h(V) = 1$ , even valuation at the inert primes and the unit  $\eta$ , and not, as the proof says in terms, by Hasse. An earlier version of this paper stated it as verified and not proved, having missed that  $3R$  is the maximal order. Theorem 1 does not—it is proved—and the constructions of Section 6 are exhibited rather than predicted, so nothing in Section 6 rests on the criterion either.

The choice of  $R$  is a choice. Taking  $|j| \leq J$  instead of  $|j| \leq 2$  in the generating family gives, for each  $J$ , a rank-4 lattice, and these nest:  $30 \subset 42 \subset 54 \subset \dots$ . The index over ours is  $9^{J-2}$ , checked for  $J \leq 5$ , and the unit vectors are exactly the  $6(2J + 1)$  generators, checked for  $J \leq 3$ . Neither is proved here for all  $J$ ; counting generators does not by itself rule out a further unit vector, and nothing below depends on the general statement. Heule's 553-vertex large part lives in the 42-vector member. Everything here is about the 30-vector one, which is what the record and every construction in Section 6 use; the law is not claimed for the finer members, and the criterion would have to be recomputed on each.

Nothing here reaches the rotations of [9] that are not spindles, and we checked rather than assumed it. Their four are  $\rho u$  for a unit vector  $u$  of  $R$ ; the whole family  $\{\rho u\}$  is 30 rotations, of which six—the sixth roots of unity—fix  $R$  and so give back  $\theta_4$ 's own joint lattice, and exactly two are spindles,  $\rho$  and  $-\rho$ . Every one of the 30 has a rank-8 joint lattice with 126, 108 or 90 unit vectors. Run against all 30, the criterion of Section 2 kills *none*, and the stronger periodic filter of Section 7 kills none either, degenerating at  $m = 4$  and  $m = 8$  alike. So both tools are blind outside the spindle family: against the integer spindles the law removes three  $a$  in four and the periodic filter eleven of the sixteen survivors, and against  $\{\rho u\}$  they remove nothing at all. Whatever decides that family is not in this paper.

No claim is made about the record. The bound 509 is unbeaten, and nothing here brings it closer. What is settled is an obstruction—no spindle of rank 8 with  $4 \nmid a$  can be used at all, the rank-4 exceptions of Proposition 4 being untouched—together with a proved criterion for occupancy, eleven further exclusions and seven constructions. What is not settled is the classification those would have to add up to: rotations outside the spindle family are untouched.

## CODE AND DATA

Every computation quoted here is carried out by a script in the accompanying repository, <https://github.com/calinnedelcu/joining-rotations-artifacts>, whose `README.md` maps each claim to the script that reproduces it and says, claim by claim, what the archive settles and what it does not. The snapshot for this draft is commit `e1e582848570edd8dbce3b5f63e425fbde7ace43`; we quote the commit and not a tag because a tag can be moved and a hash cannot.

`scripts/audit_numbers.py` re-derives the paper’s numbers from the definitions in about three minutes, on the standard library alone, sharing no code with the rest. `scripts/check_emptyshell.py` and `check_periodic.py` validate the twenty witnesses from their JSON alone, importing nothing from the project. `scripts/check_drat.sh` verifies the eight DRAT certificates—every construction of the companion paper, the gadget Lemma 20 rests on, and Parts’ record—in about ten minutes; `drat-trim` is deliberately not vendored, since a checker obtained from the people whose proof you are checking is not a check. `verify_all.sh` runs all 36 recorded claims in parallel, a little over an hour on six cores; the last full run passed 35 of 35. `PROVENANCE.md` names the four files taken from [6], and `LICENSES.md` the licence split: MIT for the code, CC BY 4.0 for the paper, the certificates and the data.

A companion paper treating the seven constructions and the  $\theta_4$  interface bound in detail is in preparation; nothing above depends on it, and every computation it will describe is already in the archive at the commit named here.

Archived at Zenodo, [10.5281/zenodo.22855504](https://zenodo.org/record/22855504). That is the *concept* DOI: it resolves to the newest deposited version, so it stays right when the archive is updated, while the commit named above says which snapshot these numbers came from.

## ACKNOWLEDGEMENTS

The homomorphism 4-colouring of the Moser ring, which this paper’s criterion redirects at the joint lattice, is due to Philip Gibbs in the Polymath16 project; the  $\omega_t$  notation and the algebraic framework are Polymath16’s. The classification of the Moser lattice’s two geometric 4-colourings is due to Ákos Dúcz.

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